Problem 4

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Problem 1 *Use the Intermediate value theorem to find an interval in which you can guarantee that there is a solution to the equation $x^3 = x + \sin(x) + 1$.* ***EXPLAIN****. (Do not use a graphing device or calculator to solve this problem!)*

Explanation. We have

$$x^3 = x + \sin(x) + 1 \iff x^3 - x - \sin(x) - 1 = 0.$$

Define $f(x) = x^3 - x - \sin(x) - 1$.

Since:
$f(0) = (0)^3 - (0) - \sin(0) - 1 = -1$ and
$f(\pi) = \pi^3 - \pi - \sin(\pi) - 1 = \pi(\pi^2 - 1) - 1 > 3 \cdot (3^2 - 1) - 1 = 23$
We have: $-1 < 0 < 23$
and f is continuous on $[0, \pi]$, the Intermediate Value Theorem implies there is some c with $0 < c < \pi$ such that $f(c) = L = 0$, that is, $c^3 = c + \sin(c) + 1$. That c is a solution on the interval $[0, \pi]$. (Note: There are many possible intervals, e.g. $[1, 4]$, etc.)